

Predictive Forecasting of Care Load & Placement Demand

Abstract

The Unaccompanied Alien Children (UAC) Program operates in a dynamic environment where sudden changes in migration patterns can significantly impact the number of children entering federal care. Accurate forecasting of care load and placement demand is essential for proactive resource planning and operational efficiency. This project applies machine learning and time-series forecasting techniques to predict the number of children in HHS care using historical daily data. The study includes feature engineering, model development, performance evaluation, and future forecasting to support data-driven decision-making.

Introduction

The U.S. Department of Health and Human Services (HHS) is responsible for providing care and placement services for unaccompanied children transferred from Customs and Border Protection (CBP). Variations in border activity and policy changes can lead to fluctuations in the number of children requiring care.

Traditional reporting methods focus on historical data and do not provide sufficient foresight for operational planning. Predictive forecasting enables administrators to anticipate future care demand, allocate resources efficiently, and reduce overcrowding risks.

Problem Statement

Despite having access to daily operational data, the UAC Program lacks reliable short-term forecasting capabilities. This limitation creates challenges in planning shelter capacity, staffing requirements, and discharge operations.

The objectives of this project are:

- Forecast the number of children in HHS care.

- Estimate future care load trends.
- Evaluate different forecasting techniques.
- Support proactive healthcare and child welfare planning.

Dataset Description

The dataset contains daily operational statistics from the UAC Program.

Dataset Attributes

Column Name	Description
Date	Reporting date
Children apprehended and placed in CBP custody	Daily intake volume
Children in CBP custody	Active CBP care load
Children transferred out of CBP custody	Transfers into HHS care
Children in HHS Care	Active HHS care load
Children discharged from HHS Care	Successful sponsor placements

The dataset contains 1170 observations collected over multiple years.

Data Preprocessing

The following preprocessing steps were performed:

1. Converted the Date column to datetime format.
2. Sorted records chronologically.
3. Removed formatting characters such as commas from numeric fields.
4. Handled missing values using interpolation.
5. Set the Date column as the time-series index.

Feature Engineering

To improve forecasting performance, additional features were created:

Lag Features

- Lag 1 Day
- Lag 7 Days
- Lag 14 Days

Rolling Statistics

- 7-Day Rolling Mean
- 14-Day Rolling Mean

Flow-Based Indicator

Net Pressure was calculated as:

Net Pressure = Transfers Out of CBP Custody – Discharges from HHS Care

Calendar Features

- Day of Week
- Month
- Quarter

These features help capture temporal patterns and operational pressures.

Forecasting Models

Several forecasting models were implemented and compared.

Baseline Models

1. Naive Forecast
2. Moving Average Forecast

Statistical Models

1. ARIMA

2. SARIMA
3. Exponential Smoothing

Machine Learning Models

1. Random Forest Regressor
2. Gradient Boosting Regressor

The machine learning models were trained using engineered features and evaluated on a time-based test set.

Model Evaluation

The following evaluation metrics were used:

Mean Absolute Error (MAE)

Measures average prediction error.

Root Mean Squared Error (RMSE)

Penalizes larger forecasting errors.

Mean Absolute Percentage Error (MAPE)

Measures prediction accuracy as a percentage.

Forecast Accuracy

Forecast Accuracy (%) = $100 - \text{MAPE}$

Results and Analysis

The forecasting models successfully captured the trend of children in HHS care. Machine learning models demonstrated strong predictive performance due to the use of lag features and operational indicators.

Key observations:

- Historical care load exhibits clear trends over time.
- Net Pressure significantly influences future care demand.
- Random Forest and Gradient Boosting provided stable predictions.
- Forecasting can provide early warning signals for capacity planning.

The generated forecast supports short-term operational planning and resource allocation.

Future Forecasting

A future forecast was generated for the next 30 days using the trained forecasting models. The forecast can be used by decision-makers to estimate expected care demand and prepare resources in advance.

Conclusion

This project demonstrates the application of predictive analytics for forecasting care load and placement demand within the UAC Program. By leveraging historical operational data, feature engineering, and forecasting techniques, the system provides valuable insights into future care requirements.

The developed forecasting framework supports proactive planning, improves operational readiness, and enhances decision-making for healthcare and child welfare stakeholders. Future enhancements may include advanced deep learning models, real-time forecasting dashboards, and integration of external policy and migration indicators.